MSG No. 134.475 $P4_2'/nnm'$ [Type III, tetragonal]

Table 1: Wyckoff site: `2a`, site symmetry: `-4'2m'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[5,6,7,8,9,10,11,12]`

Table 2: Wyckoff site: `2b`, site symmetry: `-4'2m'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,2,3,4,13,14,15,16]`
2	$\left[\frac{1}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[5,6,7,8,9,10,11,12]`

Table 3: Wyckoff site: `4c`, site symmetry: `222.`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$	`[1,2,3,4]`
2	$\left[\frac{3}{4}, \frac{3}{4}, \frac{3}{4}\right]$	`[5,6,7,8]`
3	$\left[\frac{1}{4}, \frac{1}{4}, \frac{3}{4}\right]$	`[9,10,11,12]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{4}\right]$	`[13,14,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.2'2'`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[1,4,11,12]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[2,3,9,10]`
3	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[5,8,15,16]`
4	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[6,7,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `..2'/m'`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,5,11,15]`
2	$\left[0, \frac{1}{2}, 0\right]$	`[2,6,10,14]`
3	$\left[\frac{1}{2}, 0, 0\right]$	`[3,7,9,13]`
4	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[4,8,12,16]`

Table 6: Wyckoff site: `4f`, site symmetry: `..2'/m'`

No.	position	mapping
1	$[0, 0, 0]$	`[1,5,11,15]`
2	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[2,6,10,14]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[3,7,9,13]`

continued ...

Table 6

No.	position	mapping
4	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[4,8,12,16]`

Table 7: Wyckoff site: `4g`, site symmetry: `2.m'm'`

No.	position	mapping
1	$\left[\frac{3}{4}, \frac{1}{4}, z\right]$	`[1,4,15,16]`
2	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[2,3,13,14]`
3	$\left[\frac{1}{4}, \frac{3}{4}, -z\right]$	`[5,8,11,12]`
4	$\left[\frac{1}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[6,7,9,10]`

Table 8: Wyckoff site: `8h`, site symmetry: `2..`

No.	position	mapping
1	$\left[\frac{1}{4}, \frac{1}{4}, z\right]$	`[1,4]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}-z\right]$	`[2,3]`
3	$\left[\frac{3}{4}, \frac{3}{4}, -z\right]$	`[5,8]`
4	$\left[\frac{3}{4}, \frac{3}{4}, z+\frac{1}{2}\right]$	`[6,7]`
5	$\left[\frac{1}{4}, \frac{1}{4}, z+\frac{1}{2}\right]$	`[9,10]`
6	$\left[\frac{1}{4}, \frac{1}{4}, -z\right]$	`[11,12]`
7	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}-z\right]$	`[13,14]`
8	$\left[\frac{3}{4}, \frac{3}{4}, z\right]$	`[15,16]`

Table 9: Wyckoff site: `8i`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{3}{4}\right]$	`[1,2]`
2	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{3}{4}\right]$	`[3,4]`
3	$\left[-x, \frac{3}{4}, \frac{1}{4}\right]$	`[5,6]`
4	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[7,8]`
5	$\left[\frac{1}{4}, x, \frac{1}{4}\right]$	`[9,11]`
6	$\left[\frac{1}{4}, \frac{1}{2}-x, \frac{1}{4}\right]$	`[10,12]`
7	$\left[\frac{3}{4}, -x, \frac{3}{4}\right]$	`[13,15]`
8	$\left[\frac{3}{4}, x+\frac{1}{2}, \frac{3}{4}\right]$	`[14,16]`

Table 10: Wyckoff site: `8j`, site symmetry: `.2.`

No.	position	mapping
1	$\left[x, \frac{1}{4}, \frac{1}{4}\right]$	`[1,2]`
2	$\left[\frac{1}{2}-x, \frac{1}{4}, \frac{1}{4}\right]$	`[3,4]`
3	$\left[-x, \frac{3}{4}, \frac{3}{4}\right]$	`[5,6]`
4	$\left[x+\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,8]`
5	$\left[\frac{1}{4}, x, \frac{3}{4}\right]$	`[9,11]`

continued ...

Table 10

No.	position	mapping
6	$\left[\frac{1}{4},\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[10,12]
7	$\left[\frac{3}{4},\ -x,\ \frac{1}{4}\right]$	[13,15]
8	$\left[\frac{3}{4},\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[14,16]

Table 11: Wyckoff site: 8k, site symmetry: ..2'

No.	position	mapping
1	$[x,\ x,\ 0]$	[1,11]
2	$\left[x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[2,10]
3	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{2}\right]$	[3,9]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ 0\right]$	[4,12]
5	$[-x,\ -x,\ 0]$	[5,15]
6	$\left[-x,\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[6,14]
7	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{2}\right]$	[7,13]
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ 0\right]$	[8,16]

Table 12: Wyckoff site: 8l, site symmetry: ..2'

No.	position	mapping
1	$\left[x,\ x,\ \frac{1}{2}\right]$	[1,11]
2	$\left[x,\ \frac{1}{2}-x,\ 0\right]$	[2,10]
3	$\left[\frac{1}{2}-x,\ x,\ 0\right]$	[3,9]
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[4,12]
5	$\left[-x,\ -x,\ \frac{1}{2}\right]$	[5,15]
6	$\left[-x,\ x+\frac{1}{2},\ 0\right]$	[6,14]
7	$\left[x+\frac{1}{2},\ -x,\ 0\right]$	[7,13]
8	$\left[x+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[8,16]

Table 13: Wyckoff site: 8m, site symmetry: ..m'

No.	position	mapping
1	$[x,\ -x,\ z]$	[1,15]
2	$\left[x,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[2,14]
3	$\left[\frac{1}{2}-x,\ -x,\ \frac{1}{2}-z\right]$	[3,13]
4	$\left[\frac{1}{2}-x,\ x+\frac{1}{2},\ z\right]$	[4,16]
5	$[-x,\ x,\ -z]$	[5,11]
6	$\left[-x,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	[6,10]
7	$\left[x+\frac{1}{2},\ x,\ z+\frac{1}{2}\right]$	[7,9]
8	$\left[x+\frac{1}{2},\ \frac{1}{2}-x,\ -z\right]$	[8,12]

Table 14: Wyckoff site: `16n`, site symmetry: `1`

No.	position	mapping
1	$[x,\ y,\ z]$	`[1]`
2	$\left[x,\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	`[2]`
3	$\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$	`[3]`
4	$\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z\right]$	`[4]`
5	$[-x,\ -y,\ -z]$	`[5]`
6	$\left[-x,\ y+\frac{1}{2},\ z+\frac{1}{2}\right]$	`[6]`
7	$\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$	`[7]`
8	$\left[x+\frac{1}{2},\ y+\frac{1}{2},\ -z\right]$	`[8]`
9	$\left[\frac{1}{2}-y,\ x,\ z+\frac{1}{2}\right]$	`[9]`
10	$\left[y,\ \frac{1}{2}-x,\ z+\frac{1}{2}\right]$	`[10]`
11	$[y,\ x,\ -z]$	`[11]`
12	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ -z\right]$	`[12]`
13	$\left[y+\frac{1}{2},\ -x,\ \frac{1}{2}-z\right]$	`[13]`
14	$\left[-y,\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	`[14]`
15	$[-y,\ -x,\ z]$	`[15]`
16	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ z\right]$	`[16]`